

PANITHAN LERTSUNTIVIT

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SUMMARY & SKILLS

UC Berkeley Mechanical Engineering Graduate with proven experience in instructing engineering courses focused on hardware integration, microprocessor architecture, and embedded development. Hands-on experience in manufacturing, rapid prototyping, electromechanical integration, and validation testing. A proactive problem-solver eager to apply hands-on technical skills to build next-generation platforms.

Software & Design: SolidWorks, Python, AutoCAD, Onshape, LaTeX, ESP-IDF, C, LabVIEW, MATLAB
Prototyping Skills: Machining, GD&T, Oscilloscopes, 3D Printing, Laser Cutting, Wire Harnessing

PROFESSIONAL EXPERIENCE

UC Berkeley Mechanical Engineering

Berkeley, CA

Teaching Assistant | ME 100 - Electronics for the Internet of Things

August 2024 - May 2025

Teaching Assistant | ME 103 - Experimentation and Measurements

August 2025 - December 2025

- Managed engineering lab resources to ensure operational efficiency and safety.
- Led weekly theoretical discussions and supervised laboratory assignments supporting over 200 students.
- Collaborated with faculty and course staff to co-develop lab activities and assessments, ensuring seamless course delivery and alignment with theoretical learning objectives.
- Mentored students in troubleshooting live hardware and software failures, ranging from circuit continuity to algorithm implementation, resulting in the successful deployment of ESP32-based IoT systems.

National Tsing Hua University

Hsinchu, Taiwan

Research Intern | Yang Lab for Biomechanics

June 2025 - August 2025

- Supported research on cuttlefish ink flow patterns. Developed an apparatus that recreated cuttlefish ink flows with variable parameters, utilizing SolidWorks to model a functional prototype that integrated mechanical parts and electrical components.
- Created a comprehensive Bill of Materials, sourcing compatible parts from accessible industrial vendors, and delivered a validated CAD model designed for assembly with a fully procured hardware inventory, enabling the lab to build the apparatus and proceed to experimentation.

Irvine Valley College

Irvine, CA

Student Tutor

June 2022 - July 2023

- Provided 1-on-1 tutoring in Mathematics, Chemistry, and Physics, deconstructing complex theoretical topics into intuitive concepts to significantly improve student academic performance.
- Mentored students in Computer-Aided Design (CAD), teaching efficient techniques to model complex geometries and troubleshoot design constraints in AutoCAD and SolidWorks.
- Administered tutoring resources, overseeing appointment logistics and student communications to enhance the center's operational efficiency.

EDUCATION

University of California, Berkeley | *B.S. Mechanical Engineering* (GPA: 3.73) December 2025
Irvine Valley College | *A.S. Engineering* (GPA: 4.00) May 2023

ORGANIZATIONS

Aerospace SAE at UC Berkeley | *Director of Avionics* 2024 - 2025

- Directed the avionics team to develop electric propulsion and telemetry systems. Oversaw the fabrication of all avionic hardware for the 2025 SAE Aero Design West Competition.
- Designed and fabricated custom wire harnesses, implementing Molex Mega-Fit connectors to ensure secure electrical continuity under high-vibration flight conditions.
- Engineered and validated a thrust test rig for propeller and motor tests in static and dynamic wind conditions, utilizing load cells and data acquisition systems.

Thai American Cultural Association at UC Berkeley | *Finance Chair* 2025

- Managed and audited financial transactions for club initiatives.

PROJECTS & INTERESTS

eV/STOL – Developed an aircraft capable of vertical/short take off and landing for increased maneuverability and efficiency.

Effects of Filament Dyes on 3D Printed Structures – Conducted tensile testing on 3D printed PLA specimen using an Instron 5500.

Real-Time Plotter – Developed a wireless ESP32-based plotter using ESP-IDF and operated from LabVIEW.

Mobile Desk – Designed a portable desk that conformed to GD&T standards. Manufactured wooden components using a laser-cutter and aluminum components using lathes and mills.

Wall Climbing Vehicle – Designed a remote control vehicle capable of maneuvering on a vertical surface using SolidWorks and Onshape. Prototyped flexible laser-cut wooden geometries.

Dark Matter Star Model – Presented a reconstructed Dark Matter self-interaction model using the Tolman-Oppenheimer-Volkoff equations at the 2023 UCI Building Bridges Conference.

Fields of Interests: Robotics, Mechatronics, Mechanical Design

Hobbies: Exploring Cultural Landmarks and Establishments and Asian Gourmet Cooking